HIGHWAY ENGINEERING
TERMWORK

12.1. SUBJECT

The calculations and drawings will be prepared for the highway being designed between given A and C points on a contour map with a scale of 1/2,000. The first necessary point will be named as "A", while the second one as "C". The direction of the designed highway is going to be from point "A" to point "C".

12.1.1. GIVEN

A- Digital contour map (Scale: 1/2.000)
B- Project criteria

1.	Project velocity (V_p)	= 70 km/h
2.	Roadway width (B)	= 10 m (2 x 3.5 +2 x 1.5)
3.	Maximum grade (g_{max})	= % 8
4.	Right-of-way width	= 40 m
5.	Minimum curve length (L)	= 50 m
6.	Maximum superelevation (q_{max})	= % 8
7.	Maximum widening of the horizontal curve	(b_w) = 2.0 m
8.	Minimum vertical curve radii	= R_{CREST} = 4,000 m
		= R_{SAG} = 2,000 m

12.1.2. PHASES

1. Determination of the most propriate route (road axis) by making a zero line study between the points A-C.
2. Preparation of the draft plan and the profile.
3. Application of a superelevation with a transition curve on a horizontal curve of the route (Clothoid will be used as the transition curve).
4. Drawing the clothoid detail and the superelevation diagram, highway revolved about the centerline.
5. Drawing the precise plan and the profile.
6. Drawing the cross-sections and calculating their areas.

12.1.3. THE FILE SHOULD INCLUDE

1. Calculations (The route study, horizontal curve, transition curve, superelevation, the grade line on the profile, vertical curve, areas of cross-sections)
2. Zero line study sheet (Scale: 1/2,000)
3. Route study sheet (Scale: 1/2,000)
4. Precise axis sheet (Scale: 1/2,000
5. Clothoid detail sheet (Scale: 1/500)
6. Superelevation detail sheet (Scale; Horizontal: 1/500, Vertical: 1/10)
7. Precise plan sheet (Scale: 1/2,000)
8. Precise profile sheet (Scale; Horizontal: 1/2000, Vertical: 1/200)
9. Cross-sections (Scale: 1/100 or 1/200)

12.2 Explanations

12.2.1.1. Providing Coordinates to the Plan sheet

Before starting operations, mark the Endpoint box in Object Snap (OSNAP) command. OSNAP command can be activated by doing one of the followings:

a) Select Menu/Tools/Drafting Settings,
b) Open settings section by right click OSNAP button under the ACAD screen
c) Write OSNAP command on to the command line at the bottom of the ACAD screen, then enter.

I – Select all document on the ACAD screen (Figure 12.1). The selection can be performed by two methods.

a) Select Menu/Edit/Select All
b) Push CTRL and A together

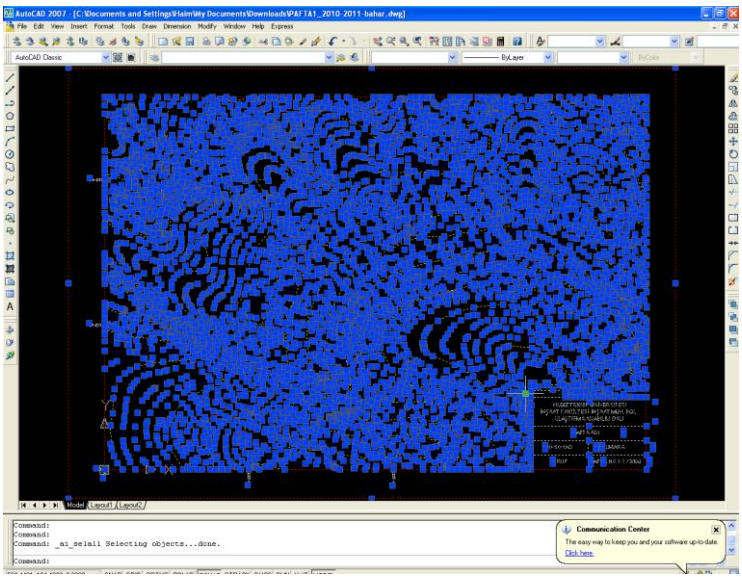
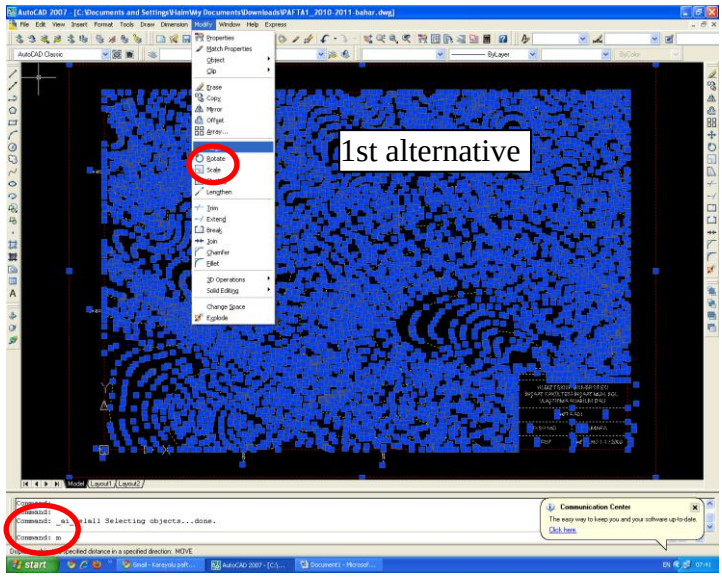


Figure 12.1 Selection of all Document

After selection of all document activate move command (figure 12.2) this can be achieved in two ways.

a) Select Menu/Modify/Move, or

b) Write m or move to the command line under the ACAD screen, than click Enter, as hown below.



12.2.2. ZERO LINE STUDY

The polygon which makes the earthwork zero on the axis will be determined by the zero line study followed between the given points. The polygon lines should not intersect or be tangent with the contour lines. Zero line will be drawn as a dashed line and with a lineweight of 0.3 mm. (Figure 12.7).

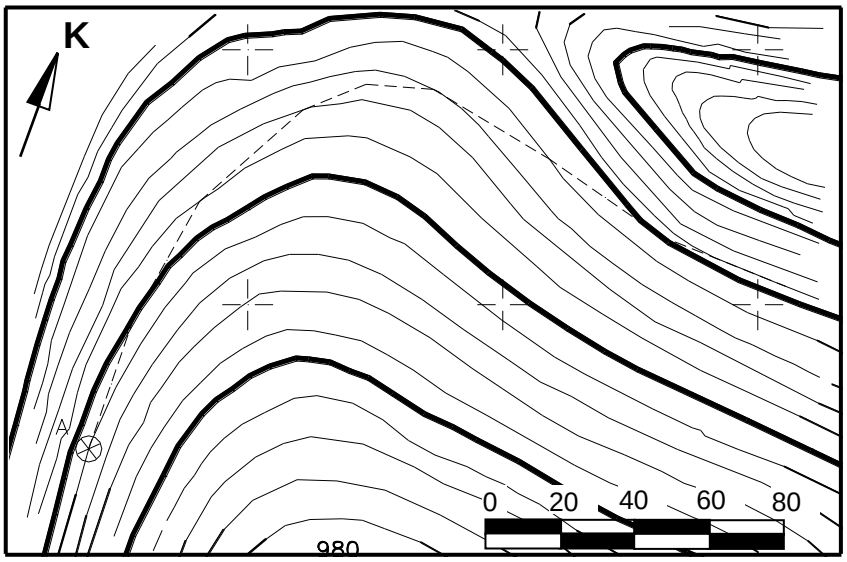


Figure 12.7 Zero Line Study

12.2.3 ROUTE STUDY

Line segments (allingments) will be placed on the obtained zero polygon and proper circle arcs (curves) will be placed between the line segments. A distance of 30 m has to be provided between the curves on the same side while 60 m has to be provided between inverse curves.

The beginning and end points of the highway and the beginning, end and the midpoint of the horizontal curves will be shown on the route. On these points, cross-section lines will be drawn vertically to the axis and with the length of the right of way (a total of 40 m, 20 m on the right and 20 m on the left). According to the direction of the highway, the point names will be written on the left side of the cross-section lines while the kilometer values will be written on the right side). The kilometer of the start point "A" will be taken as 0+000. The font of the point names and the kilometers will be "Arial" and the font size will be 3 ~ 4 mm. The cross-section linewidth will be 0.25 mm while the zero polygon, allignment and curve linewidth will be 0.3 mm. The highway platform will not be drawn on this phase. Studies are going to be folowed on the route axis. (Figure 12.8).

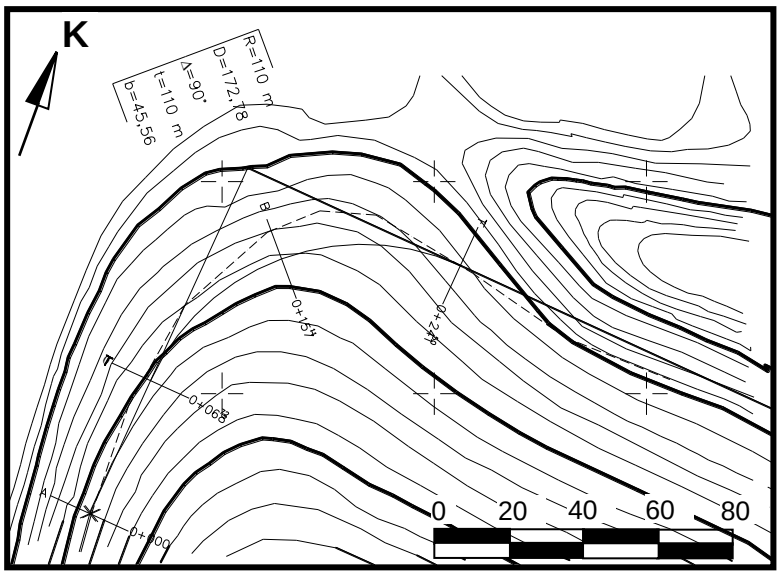


Figure 12.8 Route Study

12.2.4. PREPERATION OF THE TEMPORARY PLAN AND PROFILE

To have an idea about the earthwork amount a temporary profile will be drawn after the route study phase. For this reason, a temporary plan will be prepared first. On this plan also the highway platform will not be drawn, the study will be followed on only the axis. On the temporary plan, cross-sections will be added for each point that the route axis intersects with the contour lines, each hectometer (each 100 meters, such as H1, H2, H3, etc.), and if the distance between the consecutive cross-sections are longer than 25 m on an allignment or 20 m on a curve.

To prepare the temporary profile, the black elevations on the center of each cross-section should be known. If these points intersect on a contour line, the black line can be taken as the elevation of this contour line, directly. Else, they have to be calculated using an interpolation. Two decimal values will be taken on the elevation value found on interpolation.

By using the black elevations, the temporary profile will be prepared. By connecting the black elevations, the black line will be obtained. In this phase, preparing only the "CROSS SECTION NAME, RED ELEVATION, BLACK ELEVATION, GRADE" rows and the kilometer row in the format of (0+000) will be sufficient. As the start and end cross-sections of the highway is connected with a line, the earthwork amount of the highway will be evaluated. If the cut and fill heights are more than 10 m, they have to be decreased. For this reason, first of all, you have to return to the temporary plan and try to make changes on the route. If this is not possible, two or more red lines having different grades will be used. In this case, the algebraic difference between the grades of the red lines have to be greater than 0.5%.

12.2.5. PRECISE AXIS CALCULATIONS

The length of the route calculated during route study is an approximate value. Therefore the plan obtained is not precise but temporary and if it is necessary some arrangements can be performed. In order to maket he study more realistic precise axis study will be performed. Precise axis calculation is calculation of the route length by using analytical geometry.

Prepare the precise axis calculation sheet by transferring the allignments on the route study sheet to the map without grid and without contour lines and without the arrow directed North. Close unnecessary layers. Read the beginning adn end km of the route and km of the intersection point. The calculations should be performed in meters as unit. On the coordinate axis Y is the Horizontal and X is the Vertical . According to that the coordinates of the critical points should be written.

Also, the angles between that the allingments and the horizontal and vertical axis and curve angles should be shown symbolically. The values of the angles and lentgh of the allingments should be written on the map somewhere empty. The critical cross sections and km values (beginning and end points of the curve and allignmen) should be marked on allignments. Line weights should be similar to the route study sheet. Than precise plan drawing can be performed.

12.2.6. PRECISE AXIS CALCULATIONS

The road width and the banquets will be determined by drawing parallel lines on each side of the route axis (Road axis will be drawn as the dashed line while the banquet and the platform borders will be drawn as continous lines with the lineweight of 0,5 mm). On the precise plan, in addition to the cross-sections on the temporary plan, cross-sections will be drawn on TS, SC, CS, ST points of the transition curve, if a ramp is being applied, K_1 and K_2 points.

After all these cross-section lines are drawn, if some cross sections are intersecting with each other or are too close to each other (following each other with a distance less than 5 m), some of them will be cancelled according to their importance. The importance is the start and end points of the highway, horizontal curve, start, end and midpoints, clothoid start and end points (for each side of the curve), ramp start and end points, hectometers, and others, respectively. The cross-sections, except the special points will be enumerated starting from 1, to the end of the highway. All numbers in respect to this will be written on "Arial" font and 3 ~ 4 mm length.

On horizontal curves, curve radius (R), angle of deflection (Δ), tangent length (t), curve length (L) and external distance (e) will be written on the outside the curve according to the direction of the highway, radially and one under the other. On curves having a clothoid, A parameter and L_t length should be written as well.

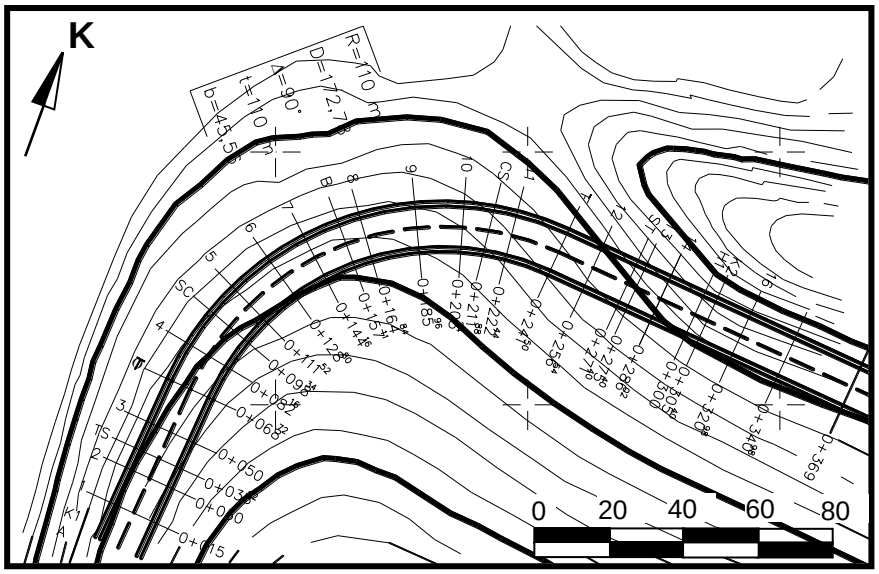


Figure 12.9 Plan

12.2.7. PROFILE

Profile is the section taken through the highway axis. The horizontal scale of the profile will be taken as 1/2,000 and the vertical scale as 1/200. To be able to place the axis superelevations on the

profile, taking the lowest superelevation on the route into account, 10 m below this value will be taken as the comparison superelevation.

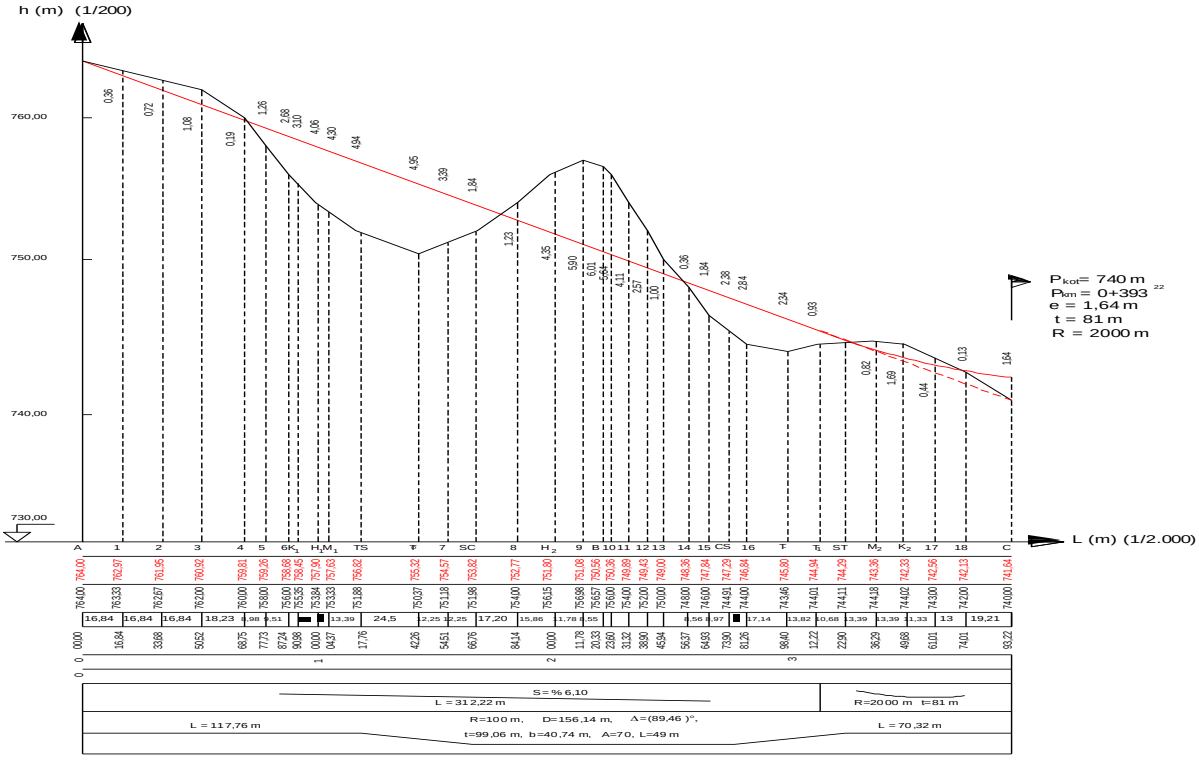


Figure 12.10 Profile

The land elevation on the axis of each cross-section of the plan will be calculated and marked on the profile as the black elevation. By connecting black elevations, the black line will be provided (0.3 mm lineweight).

By connecting the start and end points of the highway, the red line will be determined (will be drawn with a lineweight of 0.5 mm and with a red color). If more than one red lines having different grades are needed, a vertical curve will be placed on the place the red lines intersect with each other.

Each student should make vertical curve calculations within this homework. The type of the vertical curve will be circular. Minimum curve radii will be taken as R_{CREST} = 4,000 m, R_{SAG} = 2,000 m. Minimum vertical curve length is 120 m. The vertical curve will be made as:

1- If there's a need for a vertical curve in order to decrease or equalize the earthwork, it will be made. On profile there's no limit for the number of vertical curves.

2- When the vertical curve is being designed, the algebraic difference between the consecutive red line grades should fulfill the $G=g_1-g_2>\%0.5$ condition.

3- Even if the earthwork amount is too low or equalized to avoid the need for a vertical curve, a vertical curve will stil be placed. In this case, such a way will be followed:

a) If the red line has a **DESCENDING** grade, **THE END** of the highway will be connected to another red line having an **ASCENDING** grade. The grade of this red line will be taken as the last digit of the student number. In other words, if the last digit of the student number is **n**, the grade will be g_2=**%n** or g_2=**0.0n**. If the last digit of the student numbe is **0 (zero) n=%10** will be taken to continue calculations.

b) If the red line has a **ASCENDING** grade, **THE END** of the highway will be connected to another red line having an **DESCENDING** grade. The grade of this red line will be taken as the last digit of the student number. In other words, if the last digit of the student number is **n**, the grade will be g_2=**%n** or g_2=**0.0n**. If the last digit of the student numbe is **0 (zero) n=%10** will be taken to continue calculations.

A grade change sign (flag) will be placed on the places where the grade changes on the profile. At the right side of the flag, kilometer (P_{km}), elevation (P_{kot}), bisector height (e), tangent length (t) and vertical curve radius (R) values will be written from top to bottom.

The red elevations will be calculated for each cross-section. The difference between the red and the black elevations will be written under the red line on the cut, over the red line on the fill with a red color. Under the profile, the rows will be placed as seen on the figure below. The fonts on the table will be "Arial" and 3 mm height. The red elevations will be written on red color. The headings of the profile will not be included.

The header of the horizontal axis will be **l (m) (1/2,000)**; the header of the vertical axis will be **h (m) (1/200)**, written in "Arial" font and 6 mm height. The lineweight of the horizontal and the vertical axis will be 0.5 mm.

On the grade row, the value of the grade will be shown as percentage and with a 2 digit precision after the point.

On the schematical plan, the direction of the curves should be accurate. On symbolic notation, curve edges should be rounded. On curves having transition curves, TS and SC points; CS and ST points should be connected with an inclined line.

CROSS SECTION NAME	0.5 cm
RED ELEVATION	1.0 cm
BLACK ELEVATION	1.0 cm
DISTANCE	0.5 cm
DISTANCE TO START	1.0 cm
HECTOMETER	0.5 cm
KİLOMETER	0.5 cm
GRADE	1.0 cm
SCHEMATICAL PLAN	1.5 cm

Figure 12.11 Profile Table

12.2.8. Clothoid and Superelevation calculations and details

Superelevation calculation will be performed for an horizontal curve selected on the route. Superelevation will be implied on the clothoid. The jerk is limited to 0.6m/s³ while calculating the length of the clothoid. Detailed drawing of the clothoid (scale 1/500) showing all measurements on it will be prepared. Detailed drawing might be at the beginning or at the end side of the curve. It is up to students. All other cross section points related to plan should be shown on the clothoid. Show clothoid members on the map symbolically write down their values on the map. Add Δ, R, A, L values of the curve. If it is necessary perform road expansion and draw constant superelevation detail and Show all necessary measurements on it. Show the superelevation percentages of the critical points on the map schematically. Draw superelevation diagram below the detailed superelevation plan and write all necessary measurements on it. Line weigth should be 0.3mm and Arial font and 3mm height. Route axis should be on the plan with axis line and it should be coloured red. Plan detail and the scales used should be shown on the right side of the diagram.

12.2.9. Cross Sections

Three cross-sections will be drawn with a scale of 1/100 or 1/200 which can be Φ and Ŧ cross sections on any curve. Write number, km value, red and black elevation on each cross section. Calculate the cross section areas by cross method and write them below the figure. Line weight will be 0.3 mm and writings will be arial font and 3 mm size. If the cut and fill exceeds the allowable with draw retaining Wall. You will find how to draw a retaining Wall at the and of your lecture notes.